

ACCELERATING EDGE AI With Neural Systems

Artificial intelligence (AI) and machine learning (ML) are now considered a must to enhance solutions in every market segment with their ability to continuously learn and improve the experience and the output. As these solutions become popular, there is an increasing need for secure learning and customisation. However, the cost of training the AI for application-specific tasks is the primary challenge across the market



NANDAN NAYAMPALLY is Chief Marketing Officer at BrainChip

The challenge in the market for AI to get more scaled is just the cost of doing AI. If you look at GPT-3, a single training session costs weeks and \$6 million! Therefore, every time a new model or a change is presented, again the count starts from zero and the current model needs to be retrained, which is quite expensive. Variety of things affect the training of these models, such as drift, but the need to retrain for customisation would certainly help in managing retraining cost.

Consider the manufacturing industry, which is projecting losses of about \$50 billion per year because of preventable maintenance issues. There is downtime cost because the maintenance wasn't done in time. Another staggering statistic shows the loss of productivity in the US due to people not coming to work because

the cost of preventable chronic diseases is \$1.1 trillion. That is just the productivity loss, not the cost of healthcare.

This could have been substantially reduced by more capable and cost-effective monitoring through digital health. Therefore, a need of real-time AI close to the device to help learn, predict, and correct issues and proactively schedule maintenance is duly necessary.

AI is fueled by data, so one needs to consider the amount of data that is being generated by all the devices. Let us consider just one car. A single car generates data that ranges over a terabyte per day. Imagine the cost of actually learning from it. Imagine how many cars are there generating that kind of data? What does it do to the network? While not all of that data is going to the cloud, a lot of that data is being stored somewhere.

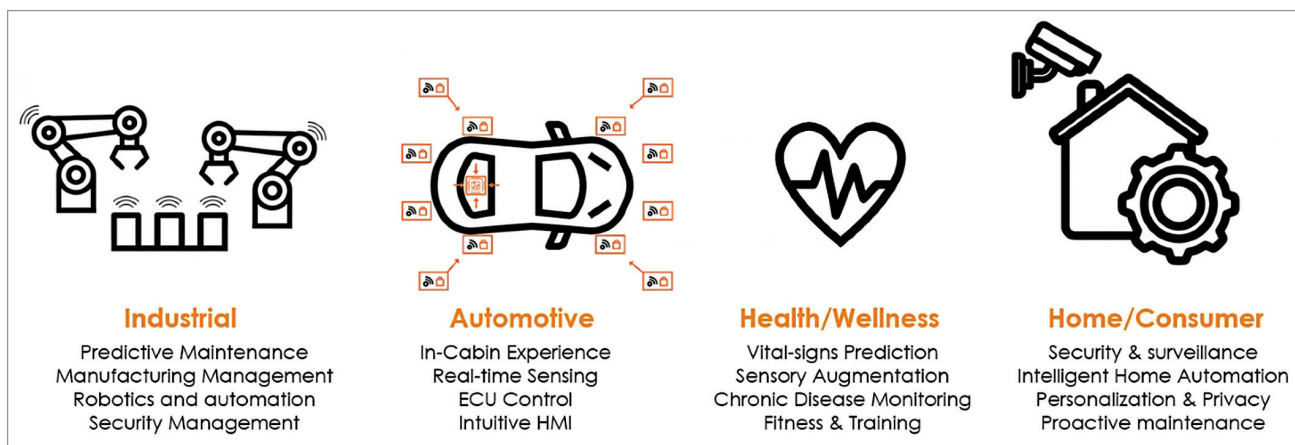


Fig. 1: Various market sectors and how neuromorphic systems and AI can be utilised in the respective industry (Credit: BrainChip)